

AI for Interpretable Heart Disease Prediction

Clinical Prediction Report

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Prediction Summary

■ No Heart Disease (Low Risk)

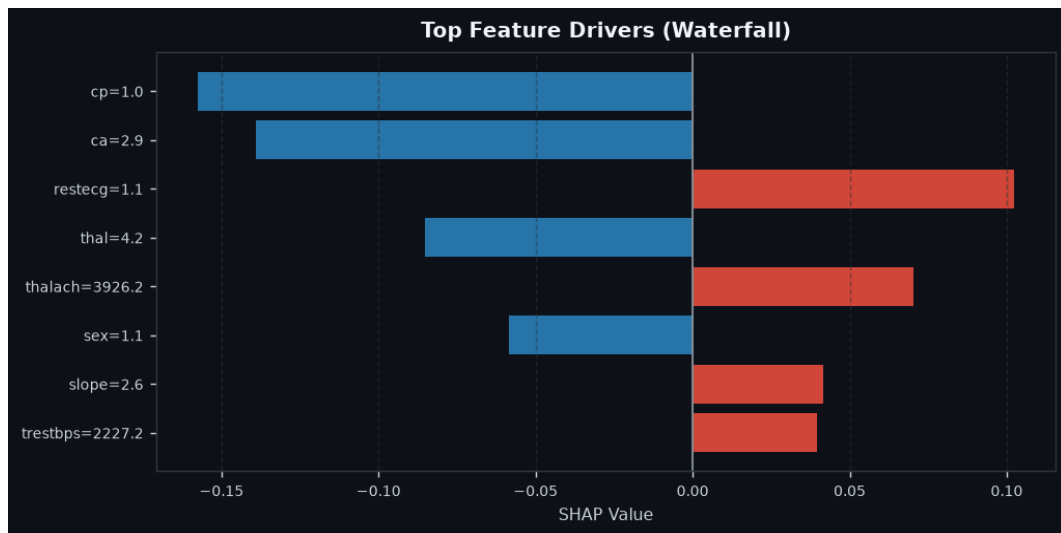
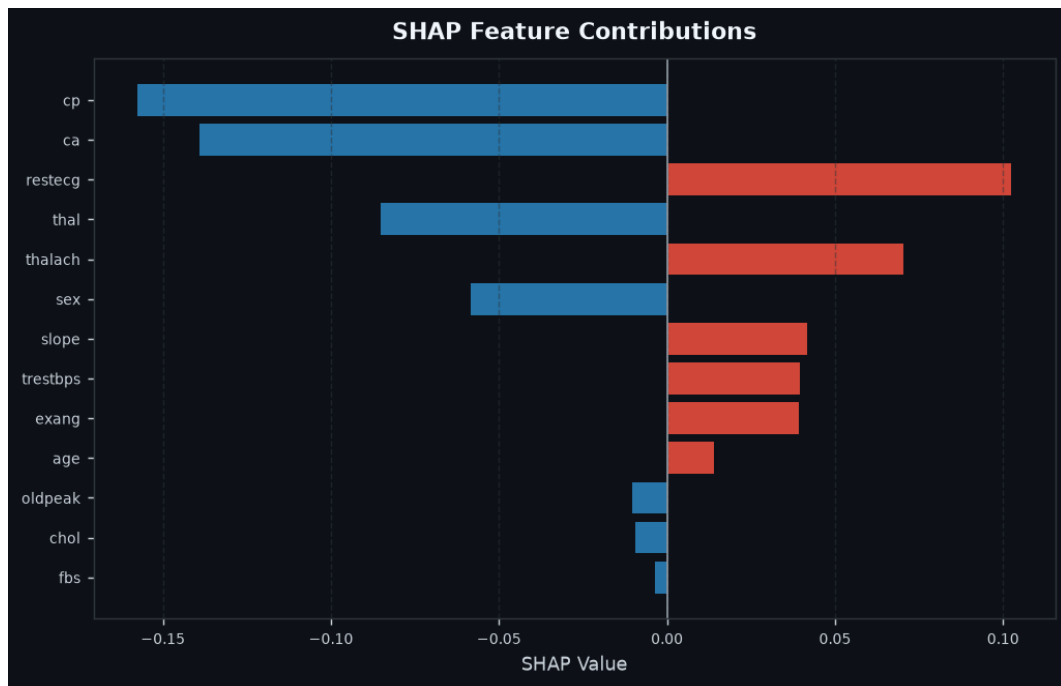
Model	Disease Probability	Contribution
Random Forest	17.8%	33%
SVM (RBF)	40.2%	33%
ANN (MLP)	39.3%	33%
Ensemble (Avg)	32.4%	Final Decision

Patient Clinical Features

Feature	Value	SHAP Impact	Direction
Chest Pain Type (0–3)	0.0	-0.1576	↓ Risk
Coronary Arteries Coloured (0–4)	2.0	-0.1391	↓ Risk
Resting ECG (0–2)	1.0	+0.1025	↑ Risk
Thalassemia (0–3)	3.0	-0.0852	↓ Risk
Max Heart Rate Achieved	168.0	+0.0703	↑ Risk
Sex (1=Male, 0=Female)	1.0	-0.0584	↓ Risk
Slope of ST Segment (0–2)	2.0	+0.0417	↑ Risk
Resting Blood Pressure (mmHg)	125.0	+0.0395	↑ Risk
Exercise Induced Angina (1=Yes)	0.0	+0.0393	↑ Risk
Age (years)	52.0	+0.0140	↑ Risk
ST Depression (oldpeak)	1.0	-0.0103	↓ Risk
Cholesterol (mg/dl)	212.0	-0.0094	↓ Risk
Fasting Blood Sugar > 120 (1=Yes)	1.0	-0.0036	↓ Risk

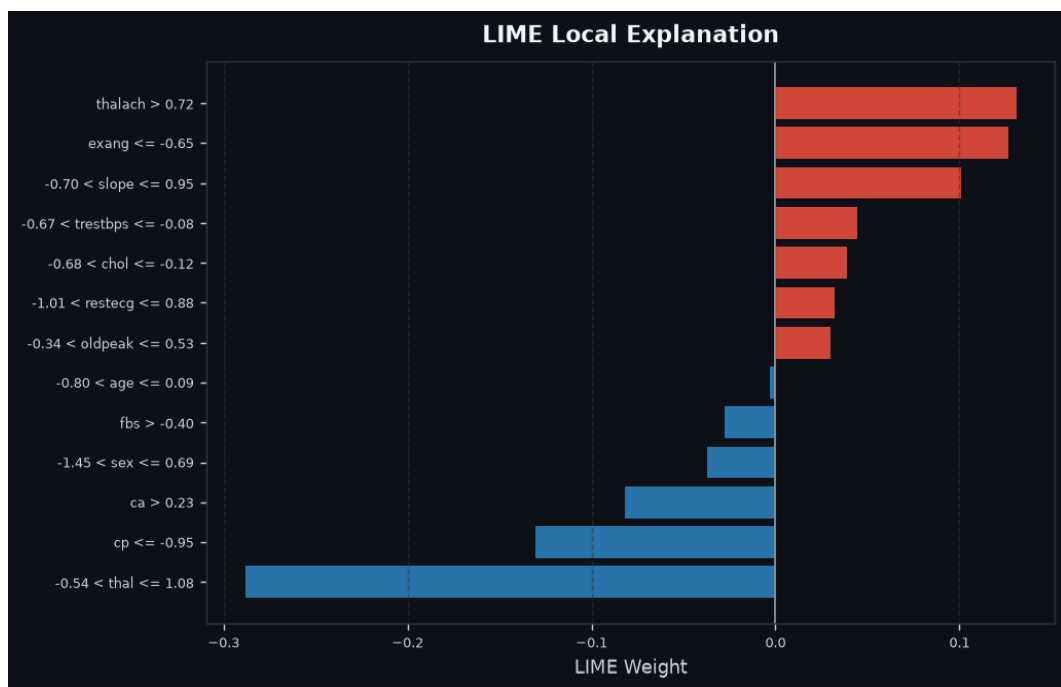
SHAP Explainability

SHAP (SHapley Additive exPlanations) shows how each clinical feature contributed to this prediction. Red bars push toward disease; blue bars push away from disease.



LIME Explainability

LIME (Local Interpretable Model-Agnostic Explanations) fits a linear model around this patient's neighbourhood to explain the prediction locally. Positive weights push toward disease; negative weights push away.



■ **DISCLAIMER:** This report is generated by an AI research system for educational and research purposes only. It is NOT a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified physician.